

Appendix C

Summary of Formulas

Chapter 2

API Gravity/Specific Gravity

$$\text{Specific gravity (Sg)} = 141.5 / (131.5 + \text{API}) \quad (2.1)$$

$$\text{API} = 141.5 / \text{Sg} - 131.5 \quad (2.2)$$

Specific Gravity Versus Temperature

$$S_T = S_{60} - a(T - 60) \quad (2.3)$$

where

S_T = Specific gravity at temperature T

S_{60} = Specific gravity at 60° F

T = Temperature, °F

a = A constant that depends on the liquid

Specific Gravity Blending

$$S_b = \frac{(Q_1 \times S_1) + (Q_2 \times S_2) + (Q_3 \times S_3) + \dots}{Q_1 + Q_2 + Q_3 + \dots} \quad (2.4)$$

where

S_b =Specific gravity of the blended liquid

Q_1, Q_2, Q_3 , etc.=Volume of each component

S_1, S_2, S_3 , etc.=Specific gravity of each component

Viscosity Conversion

$$\text{Centistokes} = 0.226(\text{SSU}) - 195/(\text{SSU}) \quad \text{for } 32 \leq \text{SSU} \leq 100 \quad (2.8)$$

$$\text{Centistokes} = 0.220(\text{SSU}) - 135/(\text{SSU}) \quad \text{for } \text{SSU} > 100 \quad (2.9)$$

$$\text{Centistokes} = 2.24(\text{SSF}) - 184/(\text{SSF}) \quad \text{for } 25 < \text{SSF} = 40 \quad (2.10)$$

$$\text{Centistokes} = 2.16(\text{SSF}) - 60/(\text{SSF}) \quad \text{for } \text{SSF} > 40 \quad (2.11)$$

Viscosity Versus Temperature

$$\text{Log Log}(Z) = A - B \text{Log}(T) \quad (2.15)$$

where

Log=logarithm to base 10

Z depends on the viscosity of the liquid

v =viscosity of liquid, cSt

T=Absolute temperature, °R or K

A and B are constants that depend on the specific liquid. The variable Z is defined as follows:

$$Z = (v + 0.7 + C - D) \quad (2.16)$$

$$C = \exp[-1.14883 - 2.65868(v)] \quad (2.17)$$

$$D = \exp[-0.0038138 - 12.5645(v)] \quad (2.18)$$

Viscosity Blending

$$\sqrt{V_b} = \frac{Q_1 + Q_2 + Q_3 + \dots}{(Q_1/\sqrt{V_1}) + (Q_2/\sqrt{V_2}) + (Q_3/\sqrt{V_3})} \quad (2.21)$$

where

V_b =Viscosity of blend, SSU

Q_1, Q_2, Q_3 , etc.=Volumes of each component

V_1, V_2, V_3 , etc.=Viscosity of each component, SSU

$$H = 40.073 - 46.414 \text{Log}_{10} \text{Log}_{10}(V+B) \quad (2.22)$$

$$B = 0.931(1.72)^v \text{ for } 0.2 < V < 1.5 \quad (2.23)$$

$$B = 0.6 \text{ for } V \geq 1.5 \quad (2.24)$$

$$H_m = [H_1(\text{pct}_1) + H_2(\text{pct}_2) + H_3(\text{pct}_3) + \dots]/100 \quad (2.25)$$

where

H, H1, H2, etc.=Blending Index of liquids

Hm=Blending Index of mixture

B=Constant in Blending Index equation

V=Viscosity, cSt

pct1, pct2, etc.=Percentage of liquids 1, 2, etc., in blended mixture

Bulk Modulus

Adiabatic bulk modulus:

$$K_a = A + B(P) - C(T)^{1/2} - D(API) - E(API)^2 + F(T)(API) \quad (2.27)$$

where

$$A = 1.286 \times 10^6$$

$$B = 13.55$$

$$C = 4.122 \times 10^4$$

$$D = 4.53 \times 10^3$$

$$E = 10.59$$

$$F = 3.228$$

P=Pressure, psig

T=Temperature, °R

API=API gravity of liquid

Isothermal Bulk Modulus:

$$K_i = A + B(P) - C(T)^{1/2} + D(T)^{3/2} - E(API)^{3/2} \quad (2.28)$$

where

$$A = 2.619 \times 10^6$$

$$B = 9.203$$

$$C = 1.417 \times 10^5$$

$$D = 73.05$$

$$E = 341.0$$

P=Pressure, psig

T=Temperature, °R

API=API gravity of liquid

Bernoulli's Equation

$$Z_A + P_A/\gamma + V_A^2/2g + H_P = Z_B + P_B/\gamma + V_B^2/2g + \Sigma h_L \quad (2.37)$$

where

Z_A , P_A and V_A =Elevation, pressure, and liquid velocity at point A

Z_B , P_B and V_B =Elevation, pressure, and liquid velocity at point B

γ =Specific weight of the liquid

H_p =Pump head input at point A

h_L =Head lost in friction between point A and point B

Chapter 3

Pressure and Head

$$\text{Head} = 2.31(\text{psig}) / \text{Spgr ft(English units)} \quad (3.7)$$

$$\text{Head} = 0.102(\text{kPa}) / \text{Spgr m(SI units)} \quad (3.8)$$

where

Spgr=Liquid specific gravity

Velocity of Flow

$$V = 0.0119(\text{bbl/day}) / D^2 \quad (3.10)$$

$$V = 0.4085(\text{gal/min}) / D^2 \quad (3.11)$$

$$V = 0.2859(\text{bbl/hr}) / D^2 \quad (3.12)$$

where

V=Velocity, ft/s

D=Inside diameter, in.

$$V = 353.6777(\text{m}^3/\text{hr}) / D^2 \quad (3.13)$$

where

V=Velocity, m/s

D=Inside diameter, mm

Reynolds Number

$$R = VD\rho/\mu \quad (3.14)$$

$$R = VD/v \quad (3.15)$$

where

V=Average velocity, ft/s

D=Pipe internal diameter, ft

ρ =Liquid density, slugs/ft³

μ =Absolute viscosity, lb-s/ft²

R=Reynolds number, dimensionless

v=Kinematic viscosity, ft²/s

$$R = 92.24 Q / (vD) \quad (3.16)$$

where

Q=Flow rate, bbl/day

D=Internal diameter, in.

ν =Kinematic viscosity, cSt

$$R = 3160Q/(\nu D) \quad (3.17)$$

where

Q=Flow rate, gal/min

D=Internal diameter, in.

ν =Kinematic viscosity, cSt

$$R = 353,678 Q/(\nu D) \quad (3.18)$$

where

Q=Flow rate, m³/h

D=Internal diameter, mm

ν =Kinematic viscosity, cSt

Darcy-Weisbach Equation for Head Loss

$$h = f(L/D)(V^2/2g) \quad (3.19)$$

where

h=head loss, ft of liquid

f=Darcy friction factor, dimensionless

L=Pipe length, ft

D=Pipe internal diameter, ft

V=Average liquid velocity, ft/s

g=Acceleration due to gravity, 32.2 ft/s² in English units

Darcy Friction Factor

For laminar flow, with Reynolds number $R < 2000$

$$f = 64/R \quad (3.20)$$

For turbulent flow, with Reynolds number $R > 4000$ (Colebrook-White equation)

$$1/\sqrt{f} = -2\text{Log}_{10}[(e/3.7D) + 2.51/(R\sqrt{f})] \quad (3.21)$$

where

f=Darcy friction factor, dimensionless

D=Pipe internal diameter, in.

e=Absolute pipe roughness, in.

R =Reynolds number of flow, dimensionless

$$f_i = f_d / 4 \quad (3.25)$$

where

f_i =Fanning friction factor

f_d =Darcy friction factor

Pressure Drop due to Friction

$$P_m = 0.0605 f Q^2 (Sg/D^5) \quad (3.27)$$

$$P_m = 0.2421 (Q/F)^2 (Sg/D^5) \quad (3.28)$$

where

P_m =Pressure drop due to friction, psi per mile

Q =Liquid flow rate, bbl/day

f =Darcy friction factor, dimensionless

F =Transmission factor, dimensionless

Sg =Liquid specific gravity

D =Pipe internal diameter, in.

$$F = 2 / \sqrt{f} \quad (3.29)$$

where

F =Transmission factor

f =Darcy friction factor

$$F = -4 \log_{10} [(e/3.7D) + 1.255(F/R)] \quad 3.30$$

for turbulent flow $R > 4000$

In SI units

$$P_{km} = 6.2475 \times 10^{10} f Q^2 (Sg/D^5) \quad (3.31)$$

$$P_{km} = 24.99 \times 10^{10} (Q/F)^2 (Sg/D^5) \quad (3.32)$$

where

P_{km} =Pressure drop due to friction, kPa/km

Q =Liquid flow rate, m³/hr

f =Darcy friction factor, dimensionless

F =Transmission factor, dimensionless

Sg =Liquid specific gravity

D =Pipe internal diameter, mm

Colebrook-White Equation

Modified Colebrook-White Equation:

$$F = -4 \log_{10} [(e/3.7D) + 1.4125(F/R)] \quad (3.34)$$

Hazen-Williams Equation

$$h = 4.73L(Q/C)^{1.852}/D^{4.87} \quad (3.35)$$

where

h=Head loss due to friction, ft

L=Length of pipe, ft

D=Internal diameter of pipe, ft

Q=Flow rate, ft³/s

C=Hazen-Williams coefficient or C-factor, dimensionless

$$Q = 0.1482(C)(D)^{2.63} (P_m/Sg)^{0.54} \quad (3.36)$$

where

Q=Flow rate, bbl/day

D=Pipe internal diameter, in.

P_m=Frictional pressure drop, psi/mile

Sg=Liquid specific gravity

C=Hazen-Williams C-factor

$$GPM = 6.7547 \times 10^{-3} (C)(D)^{2.63} (H_L)^{0.54} \quad (3.37)$$

where

GPM=Flow rate, gal/min

H_L=Friction loss, ft per 1000 ft of pipe

In SI units

$$Q = 9.0379 \times 10^{-8} (C)(D)^{2.63} (P_{km}/Sg)^{0.54} \quad (3.38)$$

where

Q=Flow rate, m³/hr

D=Pipe internal diameter, mm

P_{km}=Frictional pressure drop, kPa/km

Sg=Liquid specific gravity

C=Hazen-Williams C-factor

Shell-MIT Equation

$$R = 92.24(Q)/(Dv) \quad (3.39)$$

$$R_m = R/(7742) \quad (3.40)$$

where

R=Reynolds number, dimensionless

R_m=Modified Reynolds number, dimensionless

Q=Flow rate, bbl/day

D=Internal diameter, in.

ν=Kinematic viscosity, cSt

$$f=0.00207/R_m \text{ (laminar flow)} \quad (3.41)$$

$$f=0.0018+0.00662(1/R_m)^{0.355} \text{ (turbulent flow)} \quad (3.42)$$

$$P_m=0.241(fS_gQ^2)/D^5 \quad (3.43)$$

where

P_m=Frictional pressure drop, psi/mile

f=Friction factor, dimensionless

S_g=Liquid specific gravity

Q=Flow rate, bbl/day

D=Pipe internal diameter, in.

In SI units

$$P_m=6.2191 \times 10^{10} (f S_g Q^2)/D^5 \quad (3.44)$$

where

P_m=Frictional pressure drop, kPa/km

f=Friction factor, dimensionless

S_g=Liquid specific gravity

Q=Flow rate, m³/hr

D=Pipe internal diameter, mm

Miller Equation

$$Q=4.06(M) (D^5 P_m / S_g)^{0.5} \quad (3.45)$$

where

$$M=\log_{10}(D^3 S_g P_m / \text{cp}^2) + 4.35 \quad (3.46)$$

and

Q=Flow rate, bbl/day

D=Pipe internal diameter, in.

P_m=Frictional pressure drop, psi/mile

S_g=Liquid specific gravity

cp=Liquid viscosity, centipoise

In SI Units

$$Q=3.996 \times 10^{-6} (M)(D^5 P_m / Sg)^{0.5} \quad (3.47)$$

where

$$M = \text{Log}_{10}(D^3 Sg P_m / \text{cp}^2) - 0.4965 \quad (3.48)$$

and

Q =Flow rate, m^3/hr

D =Pipe internal diameter, mm

P_m =Frictional pressure drop, kPa/km

Sg =Liquid specific gravity

cp =Liquid viscosity, centipoise

$$P_m = (Q/4.06M)^2 (Sg/D^5) \quad (3.49)$$

T.R.Aude Equation

$$P_m = [Q(z^{0.104})(Sg^{0.448})/(0.871(K)(D^{2.656}))]^{1.812} \quad (3.50)$$

where

P_m =Pressure drop due to friction, psi/mile

Q =Flow rate, bbl/hr

D =Pipe internal diameter, in.

Sg =Liquid specific gravity

z =Liquid viscosity, centipoise

K =T.R.Aude K -factor, usually 0.90 to 0.95

In SI Units

$$P_m = 8.888 \times 10^8 [Q(z^{0.104})(Sg^{0.448})/(K(D^{2.656}))]^{1.812} \quad (3.51)$$

where

P_m =Frictional pressure drop, kPa/km

Sg =Liquid specific gravity

Q =Flow rate, m^3/hr

D =Pipe internal diameter, mm

z =Liquid viscosity, centipoise

K =T.R.Aude K -factor, usually 0.90 to 0.95

Head Loss in Valves and Fittings

$$h = KV^2/2g \quad (3.52)$$

where

h =Head loss due to valve or fitting, ft

K =Head loss coefficient for the valve or fitting, dimensionless

V =Velocity of liquid through valve or fitting, ft/s

g =Acceleration due to gravity, 32.2 ft/s² in English units

Gradual Enlargement

$$h=K(V_1-V_2)^2/2g \quad (3.53)$$

where V_1 and V_2 are the velocity of the liquid in the smaller-diameter and larger-diameter pipe respectively. Head loss coefficient K depends upon the diameter ratio D_1/D_2 and the different cone angle due to the enlargement.

Sudden Enlargement

$$h=(V_1-V_2)^2/2g \quad (3.54)$$

Drag Reduction

$$\text{Percentage drag reduction}=100(DP_0-DP_1)/DP_0 \quad (3.55)$$

where

DP_0 =Friction drop in pipe segment without DRA, psi

DP_1 =Friction drop in pipe segment with DRA, psi

Explicit Friction Factor Equations

Churchill Equation

This equation proposed by Stuart Churchill for friction factor was reported in *Chemical Engineering* magazine in November 1977. Unlike the Colebrook-White equation, which requires trial-and-error solution, this equation is explicit in f as indicated below.

$$f=[(8/R)^{12}+1/(A+B)^{3/2}]^{1/12}$$

where

$$A=[2.457 \text{ Log}_e(1/((7/R)^{0.9}+(0.27e/D)))]^{16}$$

$$B=(37,530/R)^{16}$$

The above equation for friction factor appears to correlate well with the Colebrook-White equation.

Swamee-Jain Equation

P.K.Swamee and A.K.Jain presented this equation in 1976 in the *Journal of the Hydraulics Division of ASCE*. It is found to be the best and easiest of all explicit equations for calculating the friction factor.

$$f=0.25/[\text{Log}_{10}(e/3.7D+5.74/R^{0.9})]^2$$

It correlates well with the Colebrook-White equation.

Chapter 4

Barlow's Equation for Internal Pressure

$$S_h=PD/2t \quad (4.1)$$

where

S_h =Hoop stress, psi

P =Internal pressure, psi

D =Pipe outside diameter, in.

t =Pipe wall thickness, in.

$$S_a=PD/4t \quad (4.2)$$

where

S_a =Axial (or longitudinal) stress, psi

Internal design pressure in a pipe in English units

$$P = \frac{2T \times S \times E \times F}{D} \quad (4.3)$$

where

P =Internal pipe design pressure, psig

D =Nominal pipe outside diameter, in.

T =Nominal pipe wall thickness, in.

S =Specified minimum yield strength (SMYS) of pipe material, psig

E =Seam joint factor, 1.0 for seamless and submerged arc welded (SAW) pipes (see [Table A.11](#) in Appendix A)

F =Design factor

In SI units

$$P = \frac{2T \times S \times E \times F}{D} \quad (4.4)$$

where

P =Pipe internal design pressure, kPa

D =Nominal pipe outside diameter, mm

T =Nominal pipe wall thickness, mm

S =Specified minimum yield strength (SMYS) of pipe material, kPa

E=Seam joint factor, 1.0 for seamless and submerged arc welded (SAW) pipes (see [Table A.11](#) in Appendix A)

F=Design factor

Line Fill Volume

In English units

$$V_L = 5.129(D)^2 \quad (4.7)$$

where

V_L =Line fill volume of pipe, bbl/mile

D=Pipe internal diameter, in.

In SI units

$$V_L = 7.855 \times 10^{-4} D^2 \quad (4.8)$$

where

V_L =Line fill volume, m³/km

D=Pipe internal diameter, mm

Chapter 5

Total Pressure Required

$$P_t = P_{\text{friction}} + P_{\text{elevation}} + P_{\text{del}} \quad (5.1)$$

where

P_t =Total pressure required at A

P_{friction} =Total friction pressure drop between A and B

$P_{\text{elevation}}$ =Elevation head between A and B

P_{del} =Required delivery pressure at B

Pump Station Discharge Pressure

$$P_d = (P_t + P_s)/2 \quad (5.3)$$

where

P_d =Pump station discharge pressure

P_s =Pump station suction pressure

Equivalent Length of Series Piping

$$L_E/(D_E)^5 = L_A/(D_A)^5 + L_B/(D_B)^5 \quad (5.7)$$

where

L_E =Equivalent length of diameter D_E

L_A, D_A =Length and diameter of pipe A

L_B, D_B =Length and diameter of pipe B

Equivalent Diameter of Parallel Piping

$$Q^2/D_E^5 = Q_{BC}^2/D_{BC}^5 \quad (5.14)$$

$$Q_{BC}^2/D_{BC}^5 = (Q - Q_{BC})^2/D_{BD}^5 \quad (5.15)$$

where

Q =Total flow through both parallel pipes.

Q_{BC} =Flow through pipe branch BC

$(Q - Q_{BC})$ =Flow through pipe branch BD

D_{BC} =Internal diameter of pipe branch BC

D_{BD} =Internal diameter of pipe branch BD

D_E =Equivalent pipe diameter to replace both parallel pipes BC and BD

Horsepower Required for Pumping

$$BHP = QP / (2449E) \quad (5.16)$$

where

Q =Flow rate, bbl/hr

P =Differential pressure, psi

E =Efficiency, expressed as a decimal value less than 1.0

$$BHP = (GPM)(H)(Spgr) / (3960E) \quad (5.17)$$

$$BHP = (GPM)P / (1714E) \quad (5.18)$$

where

GPM =Flow rate, gal/min.

H =Differential head, ft

P =Differential pressure, psi

E =Efficiency, expressed as a decimal value less than 1.0

$Spgr$ =Liquid specific gravity, dimensionless

In SI units

$$\text{Power (kW)} = \frac{Q H Spgr}{367.46(E)} \quad (5.19)$$

where

Q =Flow rate, m³/hr

H=Differential head, m

Spgr=Liquid specific gravity

E=Efficiency, expressed as a decimal value less than 1.0

$$\text{Power (kW)} = \frac{Q P}{3600(E)} \quad (5.20)$$

where

P=Pressure, kPa

Q=Flow rate, m³/hr

E=Efficiency, expressed as a decimal value less than 1.0

Chapter 6

Pump Station Discharge Pressure

$$P_D = (P_T - P_s) / N + P_s \quad (6.4)$$

where

P_D=Pump station discharge pressure

P_T=Total pressure required

P_s=Pump station suction pressure

N=Number of pump stations

Line Fill Volume

$$\text{Line fill volume} = 5.129 L(D)^2 \quad (6.5)$$

where

D=Pipe internal diameter, in.

L=Pipe length, miles

Chapter 7

Brake Horsepower

$$\text{Pump BHP} = \frac{Q H S_g}{3960(E)} \quad (7.1)$$

where

Q=Pump flow rate, gal/min

H=Pump head, ft

E=Pump efficiency, as a decimal value less than 1.0

S_g=Liquid specific gravity (for water S_g=1.0)

In SI units

$$\text{Power kW} = \frac{Q H Sg}{367.46 (E)} \quad (7.2)$$

where

Q=Pump flow rate, m³/hr

H=Pump head, m

E=Pump efficiency, as a decimal value less than 1.0

Sg=Liquid specific gravity (for water Sg=1.0)

Specific Speed

$$N_s = NQ^{1/2}/H^{3/4} \quad (7.3)$$

where

N_s=Pump specific speed

N=Pump impeller speed, RPM

Q=Flow rate or capacity, gal/min

H=Head, ft

Suction Specific Speed

$$N_{ss} = NQ^{1/2}/(NPSH_R)^{3/4} \quad (7.4)$$

where

N_{ss}=Suction specific speed

N=Pump impeller speed, RPM

Q=Flow rate or capacity, gal/min

NPSH_R=NPSH required at the BEP

Impeller Diameter Change

$$Q_2/Q_1 = D_2/D_1 \quad (7.5)$$

$$H_2/H_1 = (D_2/D_1)^2 \quad (7.6)$$

where

Q₁, Q₂=Initial and final flow rates

H₁, H₂=Initial and final heads

D₁, D₂=Initial and final impeller diameters

Impeller Speed Change

$$Q_2/Q_1 = N_2/N_1 \quad (7.7)$$

$$H_2/H_1 = (N_2/N_1)^2 \quad (7.8)$$

where

Q_1, Q_2 =Initial and final flow rates

H_1, H_2 =Initial and final heads

N_1, N_2 =Initial and final impeller speeds

Suction Piping

$$\text{Suction head} = H_s - H_{fs} \quad (7.10)$$

$$\text{Discharge head} = H_D + H_{fd} \quad (7.11)$$

where

H_s =Static suction head

H_D =Static discharge head

H_{fs} =Friction loss in suction piping

H_{fd} =Friction loss in discharge piping

NPSH

$$(P_a - P_v)(2.31/Sg) + H + E1 - E2 - h \quad (7.12)$$

where

P_a =Atmospheric pressure, psi

P_v =Liquid vapor pressure at the flowing temperature, psi

Sg =Liquid specific gravity

H =Tank head, ft

$E1$ =Elevation of tank bottom, ft

$E2$ =Elevation of pump suction, ft

h =Friction loss in suction piping, ft

Chapter 8

Control Pressure and Throttle Pressure

$$P_c = P_s + \Delta P_1 + \Delta P_2 \quad (8.1)$$

where

P_c =Case pressure in pump 2 or upstream pressure at control valve

$$P_{thr} = P_c - P_d \quad (8.2)$$

where

P_{thr} =Control valve throttle pressure

P_d =Pump station discharge pressure

Chapter 9

Thermal Conductivity

$$H=K(A)(dT/dx) \quad (9.1)$$

where

H=Heat flux perpendicular to the surface area, Btu/hr

K=Thermal conductivity, Btu/hr/ft/°F

A=Area of heat flux, ft²

dx=Thickness of solid, ft

dT=Temperature difference across the solid, °F

In SI units

$$H=K(A)(dT/dx) \quad (9.2)$$

where

H=Heat flux, W

K=Thermal conductivity, W/m/°C

A=Area of heat flux, m²

dx=Thickness of solid, m

dT=Temperature difference across the solid, °C

Heat Transfer Coefficient

$$H=U(A)(dT) \quad (9.3)$$

where

U=Overall heat transfer coefficient, Btu/hr/ft²/°F

Other symbols in Equation (9.3) are the same as in Equation (9.1). In SI units

$$H=U(A)(dT) \quad (9.4)$$

where

U=Overall heat transfer coefficient, W/m²/°C

Other symbols in Equation (9.4) are the same as in Equation (9.2).

Heat Content Balance

$$H_{in}-\Delta H+H_w=H_{out} \quad (9.6)$$

where

H_{in}=Heat content entering line segment, Btu/hr

ΔH=Heat transferred from line segment to surrounding medium (soil or air), Btu/hr

H_w=Heat content from frictional work, Btu/hr

H_{out} =Heat content leaving line segment, Btu/hr

In SI units Equation (9.6) will be the same, with each term expressed in watts instead of Btu/hr.

Logarithmic Mean Temperature

$$T_m - T_s = \frac{(T_1 - T_s) - (T_2 - T_s)}{\text{Log}_e[(T_1 - T_s)/(T_2 - T_s)]} \quad (9.7)$$

where

T_m =Logarithmic mean temperature of pipe segment, °F

T_1 =Temperature of liquid entering pipe segment, °F

T_2 =Temperature of liquid leaving pipe segment, °F

T_s =Sink temperature (soil or surrounding medium), °F

In SI units Equation (9.7) will be the same, with all temperatures expressed in °C instead of °F.

Heat Content Entering and Leaving a Pipe Segment

$$H_{in} = w(C_{pi})(T_1) \quad (9.8)$$

$$H_{out} = w(C_{po})(T_2) \quad (9.9)$$

where

H_{in} =Heat content of liquid entering pipe segment, Btu/hr

H_{out} =Heat content of liquid leaving pipe segment, Btu/hr

C_{pi} =Specific heat of liquid at inlet, Btu/lb/°F

C_{po} =Specific heat of liquid at outlet, Btu/lb/°F

w =Liquid flow rate, lb/hr

T_1 =Temperature of liquid entering pipe segment, °F

T_2 =Temperature of liquid leaving pipe segment, °F

In SI units

$$H_{in} = w(C_{pi})(T_1) \quad (9.10)$$

$$H_{out} = w(C_{po})(T_2) \quad (9.11)$$

where

H_{in} =Heat content of liquid entering pipe segment, J/s (W)

H_{out} =Heat content of liquid leaving pipe segment, J/s (W)

C_{pi} =Specific heat of liquid at inlet, kJ/kg/°C

C_{po} =Specific heat of liquid at outlet, kJ/kg/°C

w =Liquid flow rate, kg/s

T_1 =Temperature of liquid entering pipe segment, °C

T_2 =Temperature of liquid leaving pipe segment, °C

Heat Transfer: Buried Pipeline

$$H_b = 6.28(L)(T_m - T_s) / (\text{Parm1} + \text{Parm2}) \quad (9.12)$$

$$\text{Parm1} = (1/K_{\text{ins}}) \text{Log}_e(R_i/R_p) \quad (9.13)$$

$$\text{Parm2} = (1/K_s) \text{Log}_e[2S/D + ((2S/D)^2 - 1)^{1/2}] \quad (9.14)$$

where

H_b = Heat transfer, Btu/hr

T_m = Log mean temperature of pipe segment, °F

T_s = Ambient soil temperature, °F

L = Pipe segment length, ft

R_i = Pipe insulation outer radius, ft

R_p = Pipe wall outer radius, ft

K_{ins} = Thermal conductivity of insulation, Btu/hr/ft/°F

K_s = Thermal conductivity of soil, Btu/hr/ft/°F

S = Depth of cover (pipe burial depth) to pipe centerline, ft

D = Pipe outside diameter, ft

In SI units

$$H_b = 6.28(L)(T_m - T_s) / (\text{Parm1} + \text{Parm2}) \quad (9.15)$$

$$\text{Parm1} = (1/K_{\text{ins}}) \text{Log}_e(R_i/R_p) \quad (9.16)$$

$$\text{Parm2} = (1/K_s) \text{Log}_e[2S/D + ((2S/D)^2 - 1)^{1/2}] \quad (9.17)$$

where

H_b = Heat transfer, W

T_m = Log mean temperature of pipe segment, °C

T_s = Ambient soil temperature, °C

L = Pipe segment length, m

R_i = Pipe insulation outer radius, mm

R_p = Pipe wall outer radius, mm

K_{ins} = Thermal conductivity of insulation, W/m/°C

K_s = Thermal conductivity of soil, W/m/°C

S = Depth of cover (pipe burial depth) to pipe centerline, mm

D = Pipe outside diameter, mm

Heat Transfer: Above-Ground Pipeline

$$H_a = 6.28(L)(T_m - T_a) / (\text{Parm1} + \text{Parm3}) \quad (9.18)$$

$$\text{Parm3} = 1.25 / [R_i(4.8 + 0.008(T_m - T_a))] \quad (9.19)$$

$$\text{Parm1} = (1/K_{\text{ins}}) \text{Log}_e(R_i/R_p) \quad (9.20)$$

where

H_a =Heat transfer, Btu/hr

T_m =Log mean temperature of pipe segment, °F

T_a =Ambient air temperature, °F

L =Pipe segment length, ft

R_i =Pipe insulation outer radius, ft

R_p =Pipe wall outer radius, ft

K_{ins} =Thermal conductivity of insulation, Btu/hr/ft/°F

In SI units

$$H_a = 6.28(L)(T_m - T_a) / (\text{Parm3} + \text{Parm1}) \quad (9.21)$$

$$\text{Parm3} = 1.25 / [R_i(4.8 + 0.008(T_m - T_a))] \quad (9.22)$$

$$\text{Parm1} = (1/K_{ins}) \text{Log}_e(R_i/R_p) \quad (9.23)$$

where

H_a =Heat transfer, W

T_m =Log mean temperature of pipe segment, °C

T_a =Ambient air temperature, °C

L =Pipe segment length, m

R_i =Pipe insulation outer radius, mm

R_p =Pipe wall outer radius, mm

K_{ins} =Thermal conductivity of insulation, W/m/°C

Frictional Heating

$$H_w = 2545(\text{HHP}) \quad (9.24)$$

$$\text{HHP} = (1.7664 \times 10^{-4})(Q)(\text{Sg})(h_f)(L_m) \quad (9.25)$$

where

H_w =Frictional heat gained, Btu/hr

HHP=Hydraulic horsepower required for pipe friction

Q =Liquid flow rate, bbl/hr

Sg =Liquid specific gravity

h_f =Frictional head loss, ft/mile

L_m =Pipe segment length, miles

In SI units

$$H_w = 1000(\text{Power}) \quad (9.26)$$

$$\text{Power} = (0.00272)(Q)(\text{Sg})(h_f)(L_m) \quad (9.27)$$

where

H_w =Frictional heat gained, W

Power=Power required for pipe friction, kW

Q=Liquid flow rate, m³/hr

Sg=Liquid specific gravity

h_f=Frictional head loss, m/km

L_m=Pipe segment length, km

Pipe Segment Outlet Temperature

For buried pipe:

$$T_2 = (1/wC_p)[2545(HHP) - H_b + (wC_p)T_1] \quad (9.28)$$

For above-ground pipe:

$$T_2 = (1/wC_p)[2545(HHP) - H_a + (wC_p)T_1] \quad (9.29)$$

where

H_b=Heat transfer for buried pipe, Btu/hr from Equation (9.12)

H_a=Heat transfer for above-ground pipe, Btu/hr from Equation (9.18)

C_p=Average specific heat of liquid in pipe segment

In SI units

For buried pipe:

$$T_2 = (1/wC_p)[1000(\text{Power}) - H_b + (wC_p)T_1] \quad (9.30)$$

For above-ground pipe:

$$T_2 = (1/wC_p)[1000(\text{Power}) - H_a + (wC_p)T_1] \quad (9.31)$$

where

H_b=Heat transfer for buried pipe, W

H_a=Heat transfer for above-ground pipe, W

Power=Frictional power defined in Equation (9.27), kW

Chapter 10

Mass Flow

$$Q = AV \quad (10.2)$$

where

A=Cross-sectional area of flow

V=Velocity of flow

Venturi Meter

Velocity in main pipe section:

$$V_1 = C\sqrt{[2g(P_1 - P_2)/\gamma]/[(A_1/A_2)^2 - 1]} \quad (10.6)$$

Velocity in throat:

$$V_2 = C\sqrt{[2g(P_1 - P_2)/\gamma]/[1 - (A_2/A_1)^2]} \quad (10.7)$$

Volume flow rate:

$$Q = CA_1\sqrt{[2g(P_1 - P_2)/\gamma]/[(A_1/A_2)^2 - 1]} \quad (10.8)$$

$$Q = CA_1\sqrt{[2g(P_1 - P_2)/\gamma]/[(1/\beta)^4 - 1]} \quad (10.9)$$

where

P_1 =Pressure in main section of diameter D and area A_1

P_2 =Pressure in throat section of diameter d and area A_2

γ =Specific weight of liquid

C =Coefficient of discharge that depends on Reynolds number and diameters

Beta ratio $\beta=d/D$ and $A_1/A_2=(D/d)^2$

Flow Nozzle Discharge Coefficient

$$C = 0.9975 - 6.53\sqrt{(\beta/R)} \quad (10.10)$$

where

$$\beta=d/D$$

and R is the Reynolds number based on the main pipe diameter D .

Turbine Meter

$$Q_b = Q_t \times M_t \times F_t \times F_p \quad (10.11)$$

where

Q_b =Flow rate at base conditions, such as 60°F and 14.7 psi

Q_t =Measured flow rate at operating conditions, such as 80°F and 350 psi

M_t =Meter factor for correcting meter reading, based on meter calibration data

F_t =Temperature correction factor for correcting from flowing temperature to the base temperature

F_p =Pressure correction factor for correcting from flowing pressure to the base pressure

Chapter 11

Pressure rise due to sudden closure of a valve:

$$\Delta H = aV/g \quad (11.1)$$

where

a =Velocity of propagation of pressure wave, ft/s

V =Velocity of liquid flow, ft/s

g =Acceleration due to gravity, ft/s²

Pressure rise due to partial closure of a valve that reduces the velocity from V to V_1 :

$$\Delta H = a(V_1 - V)/g \quad (11.2)$$

Wave speed:

$$a = \frac{(K/\rho)^{1/2}}{[1 + C(K/E)(D/t)]^{1/2}} \quad (11.3)$$

where

a =Wave speed, ft/s

K =Bulk modulus of liquid, psi

ρ =Density of liquid, slugs/ft³

C =Restraint factor, dimensionless

D =Pipe outside diameter, in.

t =Pipe wall thickness, in.

E =Young's modulus of pipe material, psi

In SI units

$$a = \frac{(K/\rho)^{1/2}}{[1 + C(K/E)(D/t)]^{1/2}} \quad (11.4)$$

where

a =Wave speed, m/s

K =Bulk modulus of liquid, kPa

ρ =Density of liquid, kg/m³

C =Restraint factor, dimensionless

D =Pipe outside diameter, mm

t =Pipe wall thickness, mm

E =Young's modulus of pipe material, kPa

The restraint factor C depends on the type of pipe condition as follows:

Case 1: Pipe is anchored at the upstream end only

Case 2: Pipe is anchored against any axial movements

Case 3: Each pipe section is anchored with expansion joints

Restraint factor C for thin-walled elastic pipes:

$$C=1-0.5\mu \text{ for case 1} \quad (11.5)$$

$$C=1-\mu^2 \text{ for case 2} \quad (11.6)$$

$$C=1.0 \text{ for case 3} \quad (11.7)$$

where μ =Poisson's ratio for pipe material, usually in the range of 0.20 to 0.45 (for steel pipe $\mu=0.30$).

For thick-walled pipes with D/t ratio less than 25, C values are as follows:

Case 1

$$C = \frac{2t}{D} (1+\mu) + \frac{D(1-0.5\mu)}{D+t} \quad (11.8)$$

Case 2

$$C = \frac{2t}{D} (1+\mu) + \frac{D(1-\mu^2)}{D+t} \quad (11.9)$$

Case 3

$$C = \frac{2t}{D} (1+\mu) + \frac{D}{D+t} \quad (11.10)$$

Chapter 12

Pipe Material Cost

$$PMC=28.1952 L(D-t)t(Cpt) \quad (12.1)$$

where

PMC =Pipe material cost, \$

L =Pipe length, miles

D =Pipe outside diameter, in.

t=Pipe wall thickness, in.

Cpt=Pipe cost, \$/ton

In SI units

$$PMC = 0.02463 L(D-t)t(Cpt) \quad (12.2)$$

where

PMC=Pipe material cost, \$

L=Pipe length, km

D=Pipe outside diameter, mm

t=Pipe wall thickness, mm

Cpt=Pipe cost, \$/metric ton